Stress-Testing Probe-Based Calibration Beyond English: Viability Boundaries, Capacity Artifacts, and When Cross-Lingual Transfer Breaks Down

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Abstract

Representation-level correctness probes offer a promising path toward calibrating large language models without output-level heuristics, but all existing work is English-only. We present the first systematic evaluation of cross-lingual correctness probe transfer across 52 languages, three architectures (two Mixture-of-Experts, one dense), and five transfer strategies. Our findings are predominantly negative but informative. First, the depth–transferability Pareto frontier—where deeper probes improve English performance but degrade cross-lingual transfer—is architecture-dependent: it is significant only on the dense architecture (Gemma 4 31B; $\rho{=}-0.48$, $p{=}0.002$) but collapses on both MoE architectures (Qwen3-235B: $\rho{=}-0.18$, $p{=}0.12$; Llama 4 Scout: $\rho{=}-0.05$, $p{=}0.62$). Second, CaCLR, a distributional-matching objective combining BCE with MMD-based consistency loss, is inert across a three-order-of-magnitude β sweep (ΔAUROC≤ 0.002, all $p{>}0.4$). Third, control task analysis reveals that 38% of language–layer probing combinations are capacity artifacts (selectivity<0.05), rising to 72% for low-resource languages—a multilingual extension of Hewitt & Liang's (2019) framework that challenges uncritical use of cross-lingual probing. Only 18 of 51 target languages (35%) meet joint viability criteria (AUROC>0.60 and selectivity>0.05), establishing a sharp viability boundary for probe-based cross-lingual calibration. Few-shot target-language probes (25–100 examples) remain the only consistently effective path forward.

1 Introduction

Non-English languages suffer substantially worse calibration in multilingual LLMs, with expected calibration error (ECE) degrading by a factor of up to 5$\times$ compared to English Ahuja et al. (2024). Output-based calibration approaches—temperature scaling, verbalized confidence elicitation—are fundamentally limited by the structural orthogonality between a model's internal correctness direction and its verbalized confidence direction Miao and Ungar (2026). Representation-level correctness probes bypass this limitation by directly accessing the linearly decodable correctness signal in hidden states Miao et al. (2026); Burns et al. (2023).

However, all existing probe-based calibration methods—CORAL Miao et al. (2026), confidence manifold approaches Yu et al. (2025), and directional steering Miao and Ungar (2026)—have been developed and evaluated exclusively in English. Xiao et al. 2025 transferred *knowledge-boundary* probes (input-level answerability) across 8 languages, but answerability is a fundamentally different target from output-dependent correctness: the former measures whether a question is answerable at all, while the latter tracks whether the model's specific output is correct.

This paper asks: *Does the probe-based calibration research program extend to non-English languages?* Our answer is a qualified negative. We evaluate cross-lingual correctness probe transfer across 52 languages, three architectures (Qwen3-235B-A22B, Llama 4 Scout, Gemma 4 31B), and five transfer conditions (zero-shot, mean-shift, linear projection, CaCLR, few-shot). Our contributions are:

1. **Viability boundary characterization.** We establish that probe-based cross-lingual calibration is viable for only 18/51 target languages (35%), all high- or mid-resource. The depth–transferability Pareto frontier is significant only on dense architectures; MoE expert routing disrupts the consistent depth–transfer tradeoff.

2. **Capacity artifact exposure.** Extending Hewitt & Liang's 2019 control task framework to multilingual probing, we find that 38% of language–layer combinations (72% for low-resource languages) have selectivity below 0.05, meaning probe "successes" in these conditions are indistinguishable from memorization artifacts.

3. **Methodological framework.** We provide a systematic evaluation protocol—combining Pareto analysis, control tasks, CKA geometric analysis, and component ablation—that future work can apply to assess cross-lingual probing viability, along with a principled negative result for MMD-based distributional matching (CACLR).

2 Related Work

Probe-based calibration. Burns et al. 2023 and Miao et al. 2026 demonstrated that correctness is linearly decodable from hidden representations. Miao & Ungar 2026 showed that correctness and verbalized confidence directions are near-orthogonal (cosine<0.04 in English), motivating representation-level approaches. All work is English-only.

Cross-lingual transfer. Pires et al. 2019 and Wu & Dredze 2019 established that multilingual models learn shared representations enabling zero-shot cross-lingual transfer, with degradation correlating with typological distance. Linear cross-lingual alignment (MUSE Conneau et al. (2018); VecMap Artetxe et al. (2018)) operates at the word-embedding level; our linear projection extends this to sentence-level correctness representations.

Domain adaptation theory. Ben-David et al. 2010 provide the H-divergence framework predicting that transfer degrades with distribution shift. Zhao et al. 2019 showed that marginal distribution matching fails when conditional distributions $P(Y|X)$ differ across domains—directly relevant to our CaCLR negative result.

Probing methodology. Hewitt & Liang 2019 introduced control tasks to distinguish genuine linguistic knowledge from probe capacity artifacts. Voita & Titov 2020 proposed MDL-based probing. We extend control tasks to multilingual settings, revealing that capacity artifacts are dramatically amplified for low-resource languages.

Conformal prediction. Angelopoulos & Bates 2023 provide distribution-free coverage guarantees but require per-language calibration data; our approach targets zero-label transfer. These are complementary directions.

3 Method

3.1 Correctness Probing Framework

Given a model $\mathcal{M}$ with L layers, we extract the hidden representation $\mathbf{h}_\ell \in \mathbb{R}^d$ at layer ℓ for each input x. A logistic regression probe $f_\ell(\mathbf{h}) = \sigma(\mathbf{w}_\ell^\top \mathbf{h} + b_\ell)$ is trained on English data to predict $y \in \{0, 1\}$

(whether $\mathcal{M}$'s output is correct). Cross-lingual transfer is evaluated by applying f_ℓ to target-language representations without retraining.

3.2 Depth–Transferability Pareto Frontier

For each target language t and architecture, we compute the Spearman rank correlation ρ_t between normalized layer depth $d_\ell = \ell/L$ and zero-shot transfer AUROC. A significant negative ρ_t indicates a Pareto tradeoff: deeper probes improve English performance but degrade cross-lingual transfer. Global ρ aggregates across all language–layer pairs.

3.3 Transfer Strategies

Zero-shot. Apply the English-trained probe directly to target-language representations.

Mean-shift. Shift target-language representations by $\boldsymbol{\mu}_{\text{en}} - \boldsymbol{\mu}_t$ before applying the probe. Under standardized features and logistic regression, mean-shift adds a constant offset to all logits; since AUROC is rank-invariant, $\Delta\text{AUROC} \equiv 0$ mathematically.

Linear projection. Learn a rotation matrix $\mathbf{R} \in \mathbb{R}^{d \times d}$ from N parallel sentence pairs to align target-language representations to the English subspace: $\hat{\mathbf{h}}_t = \mathbf{R}\mathbf{h}_t$. We evaluate $N \in \{10, 20, 50, 100, 200\}$.

CACLR. Calibration-aware Cross-Lingual Regularization jointly optimizes:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha \cdot \mathcal{L}_{\text{SB-ECE}} + \beta \cdot \mathcal{L}_{\text{MMD}} \tag{1}$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy on English labels, $\mathcal{L}_{\text{SB-ECE}}$ is soft-binned ECE, and $\mathcal{L}_{\text{MMD}}$ is the maximum mean discrepancy (Gaussian RBF kernel, median heuristic bandwidth) between English and target-language *probe logits* (1-D). α is swept over $[0.01, 10]$; β over $[0.001, 1.0]$ (three orders of magnitude). Parallel activations come from GlobalMMLU test split, partitioned into probe_train (70%) and eval (30%) via stratified split.

Few-shot. Retrain the probe on $k \in \{25, 100\}$ target-language labeled examples.

3.4 Control Task Analysis

Following Hewitt & Liang 2019, we train control probes on permuted labels and compute selectivity $= \text{AUROC}_{\text{real}} - \text{AUROC}_{\text{control}}$. A language–layer combination with selectivity<0.05 is flagged as a capacity artifact: the probe achieves similar performance on random labels, indicating memorization rather than genuine signal detection.

3.5 CKA Geometric Analysis

We compute linear CKA Kornblith et al. (2019) between English and each target language at every probed layer. To test whether CKA has independent explanatory power for transfer degradation beyond the trivial depth confound (deeper=lower CKA $\cap$ deeper=worse transfer), we report partial correlations controlling for normalized depth.

Model	Type	ρ	p	Sig. langs
Qwen3-235B	MoE	-0.18	0.12	3/19
Llama 4 Scout	MoE	-0.05	0.62	1/19
Gemma 4 31B	Dense	-0.48	0.002	9/19

Table 1: Architecture comparison: global Spearman ρ (depth vs. zero-shot transfer AUROC) and number of languages with individually significant negative correlation. Pareto frontier is significant only on the dense architecture.

3.6 Difficulty Decomposition

We train IRT difficulty probes (Ridge regression on item difficulty scores) and project the estimated difficulty direction out of the representation space. Correctness probes retrained on residuals isolate the capability-specific component. Random-direction projection serves as control.

4 Experimental Setup

Models. (1) Qwen3-235B-A22B: 94-layer MoE (22B active parameters), INT4 (GPTQ), 24 probed layers (every 4th). (2) Llama 4 Scout: 48-layer MoE (17B active of 272B total), W4A16, 12 probed layers. (3) Gemma 4 31B Dense: 60 layers, FP16, 15 probed layers. Quantization validation (§5.11) confirms rank agreement $>96\%$ between quantized and FP16 activations.

Languages. Core evaluation: 20 languages spanning 4 high-resource (zh, es, fr, de), 13 mid-resource (ja, pt, ru, ko, nl, hi, pl, id, tr, vi, uk, ar, and 1 additional), and 3 low-resource (bn, te, sw). Extended evaluation (camera-ready): 52 languages total covering 6 additional language families.

Benchmarks. GlobalMMLU (primary), Belebele (reading comprehension), MMLU-ProX (20-language MCQA). Test split partitioned: 70% probe_train, 30% eval.

Probes. Logistic regression (primary) and 2-layer MLP (ablation). 5 random seeds per configuration; 95% bootstrap confidence intervals (1000 resamples).

5 Results

5.1 Pareto Frontier Is Architecture-Dependent

Table 1 summarizes the core finding: the depth–transferability Pareto frontier, which we hypothesized to be a general property of multilingual representations, is in fact architecture-dependent. On Gemma 4 31B Dense, 9/19 target languages show individually significant negative ρ, and the global correlation is strong ($\rho{=}-0.48$, $p{=}0.002$). On both MoE architectures, the global ρ is non-significant.

MoE routing disrupts Pareto consistency. On Qwen3-235B, only 3/19 languages (zh, es, fr—all high-resource) achieve significant negative ρ, while 6/19 languages show *reversed* (positive) ρ direction, concentrated among low-resource languages (Table 2). The pattern is more extreme on Llama 4 Scout, where only Chinese achieves significance ($\rho{=}-0.62$, $p{=}0.031$) and 8/19 languages show positive ρ (Table 3). MoE expert routing is input-dependent: different languages activate different expert subsets, disrupting the monotonic depth–transfer relationship that holds in dense architectures.

Lang	Tier	ρ	p	Dir.
zh	HIGH	-0.52	0.009	expected
es	HIGH	-0.44	0.031	expected
fr	HIGH	-0.41	0.046	expected
de	HIGH	-0.33	0.11	expected
ja	MID	-0.25	0.24	expected
pt	MID	-0.28	0.19	expected
ru	MID	-0.18	0.40	expected
ko	MID	-0.06	0.79	flat
hi	MID	-0.03	0.89	flat
tr	MID	0.02	0.93	flat
vi	MID	0.12	0.58	reversed
ar	MID	0.09	0.67	reversed
bn	LOW	0.18	0.40	reversed
te	LOW	0.22	0.30	reversed
sw	LOW	0.15	0.48	reversed
Global		-0.18	0.12	—

Table 2: Qwen3-235B Pareto ρ per language (selected). Low-resource languages show reversed direction, where deeper layers *improve* transfer—inconsistent with a clean Pareto tradeoff.

Lang	Tier	ρ	p	Sig.
zh	HIGH	-0.62	0.031	✓
es	HIGH	-0.38	0.22	
de	HIGH	-0.15	0.64	
ko	MID	0.22	0.49	
vi	MID	0.25	0.43	
bn	LOW	0.28	0.38	
te	LOW	0.35	0.26	
sw	LOW	0.31	0.33	
Global		-0.05	0.62	

Table 3: Llama 4 Scout Pareto ρ (selected languages). Only zh is significant; global ρ is effectively zero.

5.2 Zero-Shot Transfer Degrades with Resource Tier

Table 4 shows zero-shot transfer AUROC on Qwen3-235B (EN-optimal layer L72, $\sim$77% depth). English achieves 0.76; the average across 20 languages drops to 0.59. The degradation follows a clear resource-tier gradient: high-resource languages lose 8.8–10.7pp, mid-resource 12.9–20.5pp, and low-resource 22.2–25.2pp. On Llama 4 Scout (Table 5), English peaks at 0.74 (L40, $\sim$83% depth) with average cross-lingual transfer at 0.57 and only 8/20 languages exceeding 0.60.

5.3 CaCLR Is Inert

Table 6 presents the central negative result. CaCLR's best configuration achieves ΔAUROC$= 0.002$ on average—within noise—while *hurting* low-resource languages (-0.008). Mean-shift produces exactly ΔAUROC$= 0.000$, as predicted by the rank-invariance proof under standardized logistic regression.

β sweep. Table 7 shows that across three orders of magnitude ($\beta \in [0.001, 1.0]$), ΔAUROC never exceeds noise level; all paired permutation tests yield $p>0.4$. At $\beta=1.0$, the MMD penalty actively hurts ($\Delta=-0.003$).

Lang	Tier	ZS	95% CI	Δ EN
en	src	.760	[.74,.78]	—
zh	HIGH	.672	[.65,.69]	−.088
es	HIGH	.668	[.64,.69]	−.092
fr	HIGH	.662	[.64,.68]	−.098
de	HIGH	.653	[.63,.68]	−.107
ja	MID	.631	[.60,.66]	−.129
ko	MID	.602	[.57,.63]	−.158
hi	MID	.588	[.56,.62]	−.172
ar	MID	.555	[.52,.59]	−.205
bn	LOW	.538	[.50,.57]	−.222
te	LOW	.521	[.49,.55]	−.239
sw	LOW	.508	[.48,.54]	−.252

Table 4: Qwen3-235B zero-shot transfer AUROC at the EN-optimal layer (L72). Transfer degradation ranges from 8.8pp (zh) to 25.2pp (sw), monotonically with resource tier.

Lang	Tier	ZS	95% CI
en	src	.740	[.72,.76]
zh	HIGH	.648	[.62,.67]
es	HIGH	.643	[.62,.67]
de	HIGH	.628	[.60,.65]
ja	MID	.601	[.57,.63]
hi	MID	.568	[.54,.60]
bn	LOW	.518	[.49,.55]
sw	LOW	.495	[.46,.53]

Table 5: Llama 4 Scout zero-shot transfer AUROC (selected).

Component ablation. Table 8 confirms no synergy between loss components. BCE+SB-ECE provides marginal ECE improvement (−0.002); BCE+MMD provides marginal AUROC improvement (+0.002); the full combination equals the maximum of individual components, not their sum.

Why does MMD fail? The MMD consistency loss operates on 1-D probe logits. It matches marginal distributions $P(\hat{y}_{\text{en}})$ and $P(\hat{y}_t)$, but the core transfer problem is that conditional distributions $P(\text{correct}|\mathbf{h})$ differ across languages Zhao et al. (2019). In MoE architectures, different languages activate different expert subsets, making the representation-space topology language-dependent in ways that 1-D marginal matching cannot address. Linear projection succeeds (14/19 languages over CACLR) because it directly learns the subspace rotation rather than indirectly matching distributions.

5.4 Capacity Artifacts in Cross-Lingual Probing

Table 9 presents the most consequential finding for the probing methodology. Control probes trained on permuted labels achieve AUROC nearly as high as real-label probes for 72% of low-resource language–layer combinations. This means that for Swahili, Bengali, and Telugu, the entire probing pipeline produces results that are likely capacity artifacts rather than genuine correctness signal detection.

Table 10 provides per-language detail at the best layer. Low-resource languages show selectivity of 0.02–0.03: real AUROC of 0.508–0.538 against control AUROC of 0.478–0.518. This finding retroactively undermines Pareto frontier analysis for low-resource languages—their data points are artifact-suspect.

Method	HIGH	MID	LOW	Wins
Mean-shift	.000	.000	.000	0/19
CaCLR (best β)	.005	.003	$-.008$	8/19
Lin. proj (50)	.025	.015	$-.005$	14/19
Few-shot (25)	.038	.030	.022	17/19
Few-shot (100)	.072	.058	.045	19/19

Table 6: ΔAUROC vs. zero-shot by transfer method and resource tier (Qwen3-235B, EN-optimal layer). CaCLR provides negligible improvement; few-shot is the only consistently effective strategy.

β	ΔAUROC	std	ΔECE	p
0.001	0.001	0.006	0.003	0.72
0.01	0.002	0.007	0.005	0.58
0.1	0.001	0.008	0.008	0.63
1.0	-0.003	0.011	0.012	0.41
0 (BCE)	0.000	0.005	0.001	—

Table 7: CaCLR β sweep (Qwen3-235B, avg. 20 languages). No β achieves significant improvement over zero-shot.

5.5 CKA Partial Correlation Is Non-Significant

Table 11 shows that naively correlating CKA with transfer degradation produces significant pooled correlations (ρ=-0.28, p=0.008 for Qwen3; ρ=-0.35, p=0.003 for Scout). However, partial correlation controlling for normalized depth reduces these to non-significance (r=0.04, p=0.58; r=0.08, p=0.35). This is a form of Simpson's paradox: depth simultaneously drives CKA decline and transfer degradation, creating a spurious direct correlation.

Same-layer analysis. At individual layers (holding depth constant), languages with higher CKA do show less transfer degradation (ρ_{avg}=-0.42), but only 5/36 layer comparisons across both models reach significance. CKA is at best a weak between-language predictor at fixed depth, not the cross-depth explanatory variable initially hypothesized.

Subspace angles. The angle between English and target-language correctness directions increases with both depth and typological distance (Table 12). Low-resource languages already show large angles (35–40°) at shallow layers, suggesting that their representations never meaningfully align with English on the correctness dimension.

5.6 Difficulty Decomposition Is Noisy

Table 13 shows that projecting out the difficulty direction provides only a marginal average improvement of ΔAUROC= $+0.005$—far smaller than few-shot (25) at $+0.032$. The difficulty-vs-random gap is significant for only 4/10 languages, all high-resource. IRT difficulty scores themselves are noisy ($R^2 = 0.35$ on average), and the "difficulty-shared" fraction estimate is highly unstable across languages (std= 0.18), precluding a consistent decomposition narrative. The difficulty component exists but is not cleanly separable in MoE architectures.

5.7 Cross-Lingual Orthogonality Degrades for Low-Resource Languages

Table 14 shows that the correctness–confidence orthogonality reported by Miao & Ungar 2026 (cosine<0.04 in English) holds for high-resource languages but systematically degrades for low-resource languages.

Components	ΔAUROC	ΔECE	Note
BCE only	.000	.001	baseline
BCE + SB-ECE	.001	−.002	α=0.1
BCE + MMD	.002	.004	β=0.01
BCE + SB-ECE + MMD	.002	.006	no synergy

Table 8: CACLR component ablation (Qwen3-235B, avg. 20 languages).

Tier	Shallow	Mid	Deep	Art. rate
EN (source)	0.03	0.12	0.25	25%
HIGH (4)	0.03	0.09	0.18	32%
MID (12)	0.02	0.06	0.10	48%
LOW (3)	0.01	0.03	0.03	72%

Table 9: Selectivity (= $\text{AUROC}_{\text{real}} - \text{AUROC}_{\text{control}}$) by resource tier and depth bin (Qwen3-235B). Artifact rate: fraction of lang–layer combinations with selectivity<0.05.

15% of language–architecture combinations exceed cosine$= 0.15$; all are low-resource languages. This could reflect genuine structural non-orthogonality or, more likely given the control task results (§5.4), noisy direction estimates from capacity-artifact probes.

5.8 Linear Probes Transfer Better Than MLPs

Table 15 confirms that linear probes are the correct choice for cross-lingual transfer. The MLP's 2.5pp English AUROC advantage is fully offset by 4pp excess transfer degradation, resulting in worse cross-lingual performance (p=0.042). The MLP captures nonlinear, language-specific patterns (including expert-routing-dependent features in MoE architectures) that do not generalize. LR transfers significantly better in 14/19 languages (bootstrap $p<0.05$).

5.9 Linear Projection Sample Efficiency

Tables 16–17 show that linear projection with 50 parallel pairs outperforms CACLR on 14/19 languages but fails for low-resource languages. With 50 pairs, low-resource languages show negative average Δ (-0.005) with high variance (std=0.028); even at 200 pairs, Swahili, Bengali, and Telugu do not consistently improve. This corroborates the subspace angle analysis: low-resource correctness directions diverge too far from English for low-rank linear alignment to help.

5.10 Cross-Benchmark Generalization

Table 18 shows that findings generalize across benchmarks. Language rankings are highly consistent with GlobalMMLU (Belebele rank r=0.82; MMLU-ProX r=0.88; Kendall τ=0.78), and CACLR remains inert on both alternative benchmarks. The low-resource AUROC drop is consistent across all three benchmarks, confirming that the viability boundary is task-general, not a GlobalMMLU-specific artifact.

5.11 Quantization Validation

Table 19 confirms that quantization introduces a small systematic AUROC reduction (avg. -0.008 for Scout W4A16; -0.012 for Qwen3 INT4) that is larger at deeper layers but does not alter per-item rankings (rank agreement $>96\%$). The $\sim$1–2pp quantization artifact is far smaller than the 10–25pp cross-lingual transfer degradation, and does not affect any qualitative conclusion.

Lang	Tier	Sel.	Art?	Real	Ctrl
en	src	.25	×	.760	.510
zh	HIGH	.18	×	.672	.492
es	HIGH	.17	×	.668	.498
ko	MID	.09	×	.602	.512
hi	MID	.10	×	.588	.488
tr	MID	.06	∼	.571	.511
ar	MID	.05	∼	.555	.505
sw	LOW	.03	✓	.508	.478
bn	LOW	.02	✓	.538	.518
te	LOW	.02	✓	.521	.501

Table 10: Per-language selectivity at the best layer (Qwen3-235B). Low-resource languages are flagged as artifacts; their real AUROC barely exceeds control AUROC.

Model	Part. r	p	Pool ρ	Interp.
Qwen3	0.04	0.58	-0.28^{**}	depth conf.
Scout	0.08	0.35	-0.35^{**}	depth conf.

Table 11: CKA partial correlation controlling for depth. Pooled correlations are significant, but this is entirely driven by the depth confound: deeper layers simultaneously have lower CKA and worse transfer.

5.12 Extended Language Coverage

Extending to 52 languages (Table 20), we classify each target language by joint viability criteria: AUROC$>$0.60 *and* selectivity$>$0.05. Only 18/51 (35%) qualify as viable—all high- or mid-resource. 22 languages (43%) are non-viable, with artifact rates exceeding 85% for newly added low-resource languages. The viability boundary is sharp: no low-resource language passes both criteria.

6 Discussion

Viability boundary interpretation. The 35% viability rate should not be interpreted as an indictment of the probe-based calibration approach per se, but rather as a boundary condition that the field must acknowledge. For the 18 viable languages, cross-lingual probe transfer is a legitimate calibration strategy with practical value. For the remaining languages, alternative approaches are needed.

MoE vs. dense architectures. The collapse of the Pareto frontier on MoE architectures is mechanistically interpretable: input-dependent expert routing creates language-specific computational pathways that break the assumption of shared representational geometry underlying cross-lingual transfer. As MoE becomes the dominant architecture for frontier models, this finding has direct implications for the scalability of probe-based methods.

Capacity artifacts as a field-level concern. The 72% artifact rate for low-resource languages extends Hewitt & Liang's 2019 insight to a setting they did not consider. Their original work focused on English syntactic probing; we show that capacity artifacts are dramatically amplified in multilingual settings, particularly for languages with lower-quality evaluation data (GlobalMMLU translation artifacts may contribute systematic patterns that probes exploit). Any future cross-lingual probing study should include control task analysis as a mandatory validation step.

Lang	Tier	L48	L72	L88
zh	HIGH	15°	22°	35°
es	HIGH	14°	20°	32°
de	HIGH	18°	25°	38°
ko	MID	25°	35°	48°
hi	MID	28°	38°	52°
sw	LOW	38°	48°	58°
bn	LOW	35°	45°	55°
te	LOW	40°	50°	58°

Table 12: Subspace angle (degrees) between English and target-language correctness directions at representative layers (Qwen3-235B).

Lang	Tier	Diff. Δ	Rand. Δ	Gap	p
zh	HIGH	.015	$-.012$	.027	.031
es	HIGH	.013	$-.014$	.027	.028
fr	HIGH	.011	$-.016$	.027	.035
de	HIGH	.008	$-.013$	.021	.062
hi	MID	.005	$-.015$	.020	.088
ru	MID	.003	$-.018$	.021	.075
ko	MID	.002	$-.014$	.016	.145
tr	MID	.001	$-.017$	.018	.118
sw	LOW	$-.005$	$-.019$	.014	.212
bn	LOW	$-.010$	$-.016$	.006	.481

Table 13: Difficulty decomposition: ΔAUROC vs. no-projection baseline. Difficulty projection is better than random projection, but the effect is small and significant only for high-resource languages.

CKA methodology warning. The depth confound in CKA analysis (significant pooled correlation but non-significant partial correlation) is a cautionary example for the representation similarity literature. Studies correlating CKA with downstream metrics without controlling for depth may report spurious relationships driven by the shared depth confounder.

Constructive paths forward. Three directions emerge for languages outside the viability boundary: (1) **Few-shot target-language probes** (25–100 examples) consistently improve transfer even for low-resource languages (ΔAUROC$= +0.022$ at k=25 for low-resource, $+0.045$ at k=100), representing the most practical near-term solution. (2) **Multilingual probe training** on pooled high-resource languages may improve the shared correctness subspace without requiring target-language labels. (3) **Dense model selection**: when Pareto analysis is required, dense architectures should be preferred over MoE for cross-lingual probing applications.

Limitations. Our evaluation is restricted to MCQA tasks, which enable clean correctness labeling but may not represent all calibration scenarios. Quantization introduces 1–2pp AUROC degradation that we account for but cannot eliminate. The causal attribution of MoE routing to Pareto collapse is correlational (n=2 MoE vs. n=1 dense); future work should test on additional architectures. Low-resource non-viability may partly reflect GlobalMMLU translation quality rather than representational limitations.

7 Conclusion

We presented the first large-scale evaluation of cross-lingual correctness probe transfer for LLM calibration, spanning 52 languages and three architectures. Our findings establish a viability boundary: probe-

Lang	Tier	Qwen3	Scout	Avg
en	src	0.03	0.03	0.03
zh	HIGH	0.04	0.03	0.035
es	HIGH	0.03	0.04	0.035
de	HIGH	0.04	0.04	0.04
ko	MID	0.06	0.07	0.065
hi	MID	0.07	0.08	0.075
tr	MID	0.08	0.09	0.085
bn	LOW	0.12	0.14	0.13
te	LOW	0.15	0.13	0.14
sw	LOW	0.14	0.18	0.16

Table 14: Cosine similarity between correctness and verbalized confidence directions. Orthogonality holds for high-resource languages (cosine<0.05) but degrades for low-resource languages (up to 0.18), qualifying the "universal orthogonality" claim.

Metric	LR	MLP	Δ	p
EN AUROC	.760	.785	+.025	$<.001$
ZS avg (20)	.590	.575	$-.015$	.042
Degradation avg	.155	.195	+.040	.008
ECE avg (20)	.180	.220	+.040	.015

Table 15: LR vs. MLP probes (Qwen3-235B). MLP's English advantage is offset and reversed in cross-lingual transfer.

based calibration transfers reliably for only 35% of tested languages, all high- or mid-resource. The depth–transferability Pareto frontier is architecture-dependent, collapsing on MoE models. MMD-based distributional matching (CACLR) is inert across all configurations. Most critically, control task analysis reveals that 38% of cross-lingual probing results—and 72% for low-resource languages—are capacity artifacts, extending Hewitt & Liang's framework to multilingual settings as a field-level methodological warning. Few-shot target-language probes (25–100 examples) remain the most reliable path forward for languages outside the viability boundary.

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N	ΔAUROC	std	Wins	Sig.
10	.005	.032	10/19	2/19
20	.010	.025	12/19	4/19
50	.015	.020	14/19	7/19
100	.022	.014	16/19	10/19
200	.028	.010	17/19	13/19

Table 16: Linear projection ΔAUROC by number of parallel pairs (Qwen3-235B, avg. 20 languages).

Tier	ΔAUROC	std	Sig.
HIGH (4)	.025	.012	3/4
MID (13)	.015	.018	4/13
LOW (3)	$-.005$	.028	0/3

Table 17: Linear projection (50 pairs) by resource tier. Unstable and *negative* for low-resource languages.

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A Full Zero-Shot Transfer Results

Due to space constraints, Tables 4 and 5 show selected languages. Full results for all 20 core languages on both architectures are available in the supplementary material. On Qwen3-235B, the zero-shot transfer success rate (defined as AUROC$>$0.60) is 55% (11/20 languages). On Llama 4 Scout, the rate drops to 40% (8/20 languages).

B Dense Architecture (Gemma 4 31B) Details

Gemma 4 31B Dense achieves EN AUROC$=$ 0.78 (L48, 80% depth) with ZS transfer average of 0.61 across 20 languages. Selectivity artifact rate is 22%, substantially lower than both MoE architectures (38–41%). Linear projection with 50 pairs achieves ΔAUROC$=$ 0.028 on average, nearly double the MoE rate. These results confirm that dense architectures remain the favorable setting for cross-lingual probe transfer.

Bench.	EN	ZS	CaCLR Δ	Rank r
GlobalMMLU	.76	.59	.002	—
Belebele	.72	.56	.001	.82
MMLU-ProX	.74	.57	.002	.88

Table 18: Cross-benchmark generalization (Qwen3-235B). Language ranking is highly consistent (Kendall τ=0.78); CACLR is consistently inert.

Model	Q	Layer	FP16	Q	Δ
Scout	W4A16	L8	.645	.642	−.003
Scout	W4A16	L40	.740	.725	−.015
Qwen3	INT4	L16	.640	.635	−.005
Qwen3	INT4	L72	.762	.742	−.020
Qwen3	INT4	L88	.745	.723	−.022

Table 19: Quantization impact on probe AUROC. Deeper layers suffer larger degradation, but rank agreement exceeds 96%.

Category	n	Criterion	Examples
Viable	18	AUROC>.60 + sel>.05	en, zh, es, fr, ja, …
Marginal	12	AUROC>.55 + sel .03–.05	hi, tr, pl, ar, …
Non-viable	22	AUROC<.55 or sel<.03	sw, bn, te, am, …

Table 20: Viability classification across 52 languages. Only 18/51 target languages (35%) meet joint viability criteria for probe-based calibration.